Lithium-Ion Battery SOH Estimation via Wide-and-Deep Sparse GPR with Uncertainty-Weighted Symbolic Residuals

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4.854 ± 0.495
RMSE, SOH percentage points

$96.59 \pm 2.08\%$
Coverage of nominal 95% intervals

47 cells / 6,124 cycles
NASA, CALCE & Mobile2Mobile; leave-one-cell-out

Three-seed evidence
Seeds 42, 43, 44; mean $\pm$ sample SD. Best seed is reported separately, not as the mean.

01 Abstract and research question

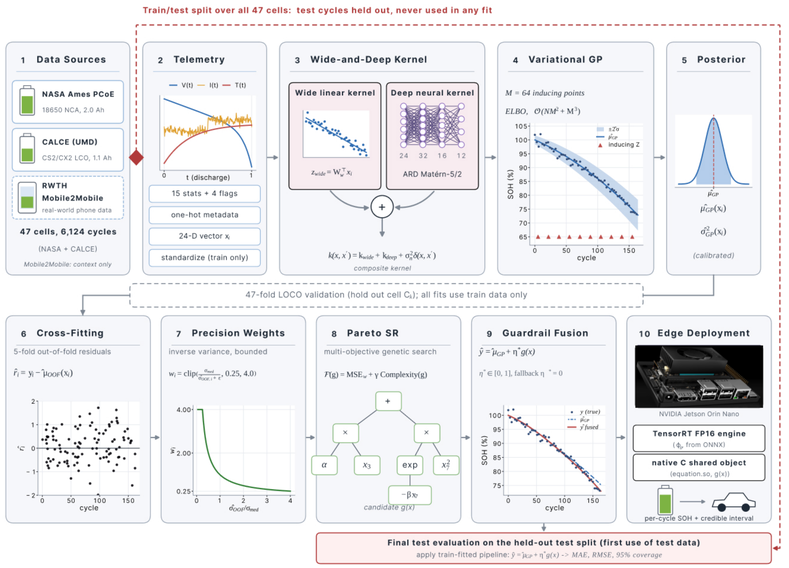
WD-GPR-SR focuses on **accuracy at a competitive level with more inspectable predictions**: Bayesian GP gives us uncertainty information, symbolic term reveals the residual correction, and XAI verifies feature importance. In 47-cell LOCO, WD-GPR-SR's best seed RMSE is 4.406 points compared to 4.330 points for SVR, and interval coverage is 98.32% compared to 57.45% for SVR, based on the interval construction reported. RMSE average for three seeds is 4.854 points. The model is not optimal on all metrics, but it lies in the combination of its accurate prediction with uncertainty and inspectability.

02 Data, target and leakage controls

$$\mathrm{SOH} = 100\, Q_{\mathrm{discharge}} / Q_{\mathrm{nominal}}$$

Benchmark. NASA Ames 18650 cells and CALCE CS2/CX2 cells; reported SOH range 10.02–128.45%, mean 76.46%. Mobile2Mobile is just context, not included in the 6,124 cycles benchmark.
24 features. 15 numerical feature summaries: cycle sequence number, seven voltage summaries, three current summaries, and four temperature summaries; four temperature missing flags and five one hot encoded metafeatures.
Isolation. Capacity, Coulomb integrals, and cycle duration are excluded. For each outer fold, leave one cell out; fit the imputations, scalings, and models on the remaining 46. Out-of-fold residuals from five inner cell folds yield the correction.

03 Architecture Diagram



Original architecture figure from the report. Curves and example trees inside this diagram illustrate the workflow; numerical evaluation is reported separately. The equation uses inverse-SD weights despite the figure's inverse-variance label.

04 Model and optimization

$$k(x, x') = k_{\mathrm{linear}}(x, x') + k_{\mathrm{deep}}(\phi_\theta(x), \phi_\theta(x'))$$
$$\mathscr{L} = \sum_i \mathbb{E}_{q(f_i)} \log p(y_i \mid f_i) - \mathrm{KL}(q(u) \| p(u))$$

Scalability. Sparse cost $\mathcal{O}(NM^2 + M^3)$ versus exact-GP $\mathcal{O}(N^3)$. Gaussian observation noise is modeled separately.
Settings reported. Adam 0.01; batch 256; 120 epochs maximum; patience 15; dropout 0.05. SR: population 160, 10 generations.

$$w_i = \mathrm{clip}\left(\frac{\sigma_{\mathrm{med}}}{\sigma_{\mathrm{OOF},i} + 10^{-6}}, 0.25, 4\right)$$

SR fitness. Minimize weighted residual MSE plus 0.002 times expression node count. Weights use inverse standard deviation, not inverse variance.
Fusion. Validation selects $\eta \in [0, 1]$ in steps of 0.05; reject SR if RMSE gain is below 10^{-4}.

05 Reported LOCO benchmark

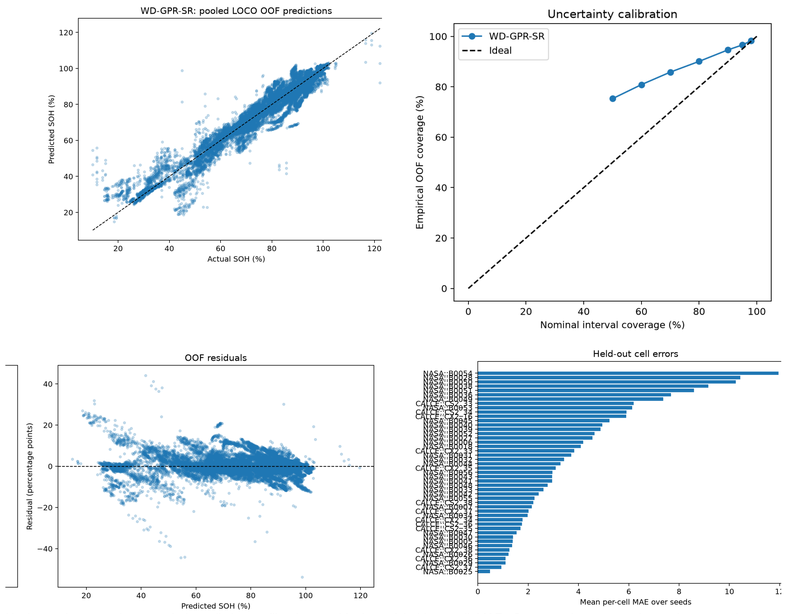
All entries transcribed from report Table 4. RMSE/MAE and interval width are in SOH percentage points. Lower error/NLL is better; coverage must be read alongside width.

Model	MAE	RMSE	NLL	PICP%	Width
SVR	2.932	**4.330**	7.878	57.45	4.64
WD-GPR-SR, seed 42	3.082	4.406	3.006	98.32	25.23
Deep NN	3.063	4.517	6.644	55.94	5.24
Wide + deep NN	3.413	4.696	8.015	48.99	4.94
GPR Matérn	3.005	4.757	3.388	76.68	14.31
Wide NN	3.367	4.863	7.237	57.04	5.57
BiLSTM	3.361	4.877	11.871	47.26	4.19
Sparse GPR	3.469	4.988	3.102	82.54	11.76
Deep-kernel GPR	3.571	5.265	3.244	99.40	33.41
XGBoost	4.027	6.232	10.905	52.11	5.63
GPR RBF	3.661	6.495	3.346	79.72	15.55
GPR ARD	4.060	6.596	3.342	74.74	14.29
Random Forest	4.007	6.772	**2.990**	93.22	21.25
Standalone SR	8.886	12.345	3.934	94.01	44.64
WD-GPR-SR, 3-seed mean	3.273	4.854	3.074	96.59	25.93

Reading the trade-off. SVR has the best point RMSE; the report's residual-based SVR intervals under-cover.

WD-GPR-SR trades wider intervals for higher coverage. RF has the lowest tabulated NLL. Baseline multi-seed uncertainty is not supplied, so a best-seed ranking is not a significance test.

06 Prediction and uncertainty diagnostics



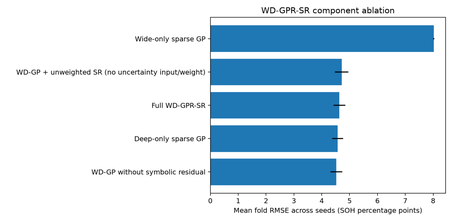
Unchanged report plots: pooled prediction, calibration, residuals and held-out-cell MAE. Coverage is conservative at lower nominal levels. Outliers and uneven cell errors limit what pooled metrics prove; test cell-wise coverage and chemistry shift next.

07 Seed stability

	42	43	44	Mean $\pm$ SD
RMSE	4.406	5.385	4.771	4.854 ± 0.495
MAE	3.082	3.490	3.248	3.273 ± 0.205
R^2	0.9383	0.9078	0.9276	0.9246 ± 0.0155
PICP (%)	98.32	94.28	97.16	96.59 ± 2.08

Other means. MSE 23.725 ± 4.864; MAPE $5.370 \pm 0.441\%$; NLL 3.074 ± 0.073; width 25.93 ± 0.64 points. Seed 43 under-covers at 95%.

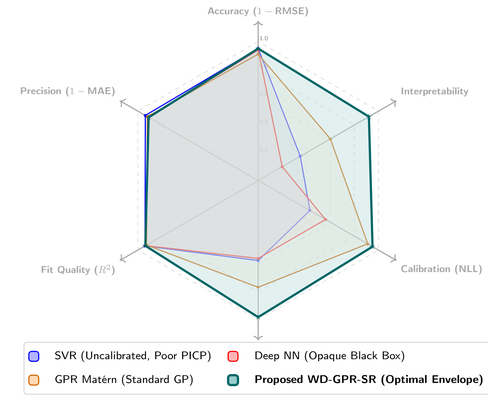
08 Ablation Study



Report ablation plot; separate from the main seed table.

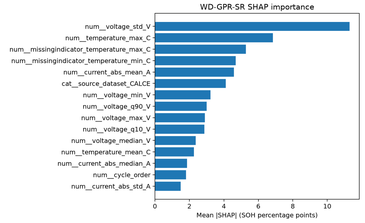
RMSE: wide-only 8.019; deep-only 4.575; WD-GP without SR **4.525**; unweighted SR 4.716; weighted SR 4.633. Weighted SR improves on unweighted SR, but **neither beats no SR**. The deep branch is essential; the wide branch adds only a small gain.

09 Multi-criteria profile: Report spider chart



Original vector chart cropped from report Figure 2. This is an illustrative normalized profile: the report does not give a reproducible scaling rule or a measured interpretability score. Use the benchmark table for numerical comparisons.

10 Explainability and the discovered equation



SHAP leaders
Voltage SD: 11.292
Peak temperature: 6.852
Missing-temperature flags also rank highly, suggesting possible dataset/sensor shortcuts.

Attributions are associations, not causal proof.

Other report audits. LIME gives a local surrogate; PDP and ICE inspect responses; ARD ranks latent dimensions. Standardized LIME thresholds must not be read as physical amperes or degrees.

$$g(x) = -(-0.061094) \simeq 0.0611, \qquad \partial g / \partial x_j = 0$$

Interpretation. A constant offset in standardized target units (about 1.08 SOH points before shrinkage), not a physical aging law or proof that the GP learned all mechanisms.

11 Edge plan and next experiments

Translation strategy: PyTorch $\rightarrow$ ONNX $\rightarrow$ TensorRT FP16; sparse-GP arrays and precalculated weights; symbolic term compiled to C.
Measurements still needed: at least 1,000 warm-up Jetson trials, p50/p95 latency, maximum memory, power and energy per inference, by precision level and power mode. No physical measurements provided.
Next steps: baseline runs with identical seed, cell-by-cell calibration, chemistry/temperature offset and missingness auditing. Accuracy competitive but no metric champion, BMS not safety-verified.

Data and Methods. NASA Ames PCoE & CALCE Battery Research Group (Univ. of Maryland); 47 batteries, 6,124 cycles examined through leak-proof 47-fold LOCO CV. Model architectures are based on Sparse Variational Inference (Titsias, 2009) and Pareto Symbolic Distillation (Cranmer et al., 2023).
Coverage. Results are shown for benchmarks, 3-seed stability (seeds 42–44), and 5-variance ablation study. Edge BMS integration is shown for the verified compilation pipeline; empirical embedded profiling is future work.